

HILLSBOROUGH, United Kingdom

WMO: 039200

Lat: 54.485N

Lon: 6.097W

Elev: 38

StdP: 100.87

Time Zone: 0.00 (EUW)

Period: 94-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-3.9	-2.2	-5.8	2.3	-1.0	-4.2	2.7	0.2	12.7	7.7	11.3	7.8	0.7	230	0.602

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.0	22.8	17.4	21.1	16.6	19.6	15.7	18.2	21.4	17.3	20.0	16.4	18.9	3.8	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.0	12.2	19.6	16.1	11.5	18.7	15.3	10.9	17.8	51.6	21.6	48.7	20.0	46.1	18.8	21.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.2	9.0	8.0	DB	-7.5	24.5	2.9	1.7	-9.6	25.7	-11.3	26.7	-13.0	27.7	-15.1	28.9	(4)
(5)			WB	-7.8	19.2	2.9	1.2	-9.9	20.1	-11.6	20.8	-13.2	21.5	-15.3	22.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.5	4.9	5.3	6.4	7.8	10.6	12.9	15.0	15.1	12.8	10.1	7.4	4.9	(6)
(7)		DBStd	4.57	2.94	2.90	2.70	2.72	2.66	2.21	2.05	2.34	2.18	2.90	2.97	3.48	(7)
(8)		HDD10.0	796	158	135	115	76	25	3	0	0	3	35	86	161	(8)
(9)		HDD18.3	3246	416	366	370	316	239	164	106	104	167	254	327	417	(9)
(10)		CDD10.0	598	1	2	3	10	44	89	154	159	87	39	9	2	(10)
(11)		CDD18.3	7	0	0	0	0	0	1	2	4	1	0	0	0	(11)
(12)		CDH23.3	31	0	0	0	0	0	6	8	17	0	0	0	0	(12)
(13)		CDH26.7	1	0	0	0	0	0	1	0	1	0	0	0	0	(13)
(14)	Wind	WSAvg	3.5	4.0	4.6	3.9	3.6	3.3	3.4	3.1	2.8	2.9	3.7	3.5	3.6	(14)
(15)	Precipitation	PrecAvg	1021	99	71	71	72	76	72	77	94	104	100	102	83	(15)
(16)		PrecMax	1261	156	176	115	131	125	117	186	152	178	170	158	160	(16)
(17)		PrecMin	756	43	16	38	32	22	32	37	16	12	46	56	16	(17)
(18)		PrecStd	131	30	38	25	28	31	26	35	39	46	42	25	42	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.0	13.5	13.9	17.9	21.2	24.0	24.4	25.4	21.1	17.7	15.1	13.9	(19)
(20)			MCWB	11.2	11.6	10.6	12.4	15.2	17.2	18.7	19.1	17.6	15.1	13.7	12.7	(20)
(21)		2%	DB	11.5	12.1	12.7	15.4	19.1	20.4	22.5	23.0	19.2	16.4	13.8	12.2	(21)
(22)			MCWB	10.0	10.3	10.1	11.6	14.1	15.3	17.2	18.1	16.0	14.3	12.5	10.9	(22)
(23)		5%	DB	10.5	11.0	11.9	13.6	17.2	18.5	20.8	21.1	18.0	15.4	12.8	11.2	(23)
(24)			MCWB	9.1	9.4	9.5	10.5	13.2	14.2	16.4	17.0	14.8	13.6	11.4	9.8	(24)
(25)		10%	DB	9.6	9.9	10.9	12.2	15.3	17.1	19.3	19.5	16.9	14.6	11.9	9.7	(25)
(26)			MCWB	8.3	8.2	8.9	9.5	11.9	13.4	15.6	16.0	14.0	12.9	10.6	8.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.2	11.7	11.8	13.7	15.8	17.6	19.7	19.9	18.1	15.6	13.8	12.9	(27)
(28)			MCDWB	12.5	13.2	12.9	16.1	19.8	22.5	23.9	23.6	20.2	17.3	14.7	13.8	(28)
(29)		2%	WB	10.2	10.6	10.9	12.1	14.6	15.8	18.0	18.6	16.8	14.7	12.7	11.0	(29)
(30)			MCDWB	11.4	11.9	12.3	14.8	18.1	19.0	21.0	22.1	18.6	16.1	13.6	12.0	(30)
(31)		5%	WB	9.2	9.6	10.0	11.0	13.4	14.9	17.0	17.6	15.6	14.0	11.7	9.9	(31)
(32)			MCDWB	10.3	10.9	11.5	13.2	16.5	17.9	19.7	20.4	17.2	15.3	12.5	11.1	(32)
(33)		10%	WB	8.5	8.4	9.1	10.0	12.3	14.0	16.1	16.7	14.5	13.1	10.8	8.6	(33)
(34)			MCDWB	9.4	9.6	10.6	11.9	14.8	16.5	18.7	19.0	16.2	14.4	11.8	9.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.0	5.9	6.5	7.5	8.0	7.2	7.3	8.0	7.6	6.3	5.6	5.2	(35)
(36)			MCDBR	5.3	6.0	7.3	8.6	10.5	9.6	10.0	10.0	9.0	6.6	5.1	5.9	(36)
(37)			MCWBR	4.9	5.5	5.2	5.5	6.6	5.7	5.7	5.6	5.8	5.1	4.6	5.5	(37)
(38)		5% WB	MCDWB	5.3	5.7	6.2	7.6	9.5	8.8	8.7	8.4	7.7	6.1	5.0	5.9	(38)
(39)			MCWBR	5.0	5.4	5.0	5.4	6.2	5.7	5.6	5.1	5.7	5.1	4.8	5.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.333	0.344	0.369	0.383	0.388	0.381	0.382	0.366	0.359	0.344	0.326	0.321	(40)	
(41)		taud	2.240	2.318	2.286	2.281	2.312	2.365	2.365	2.425	2.438	2.419	2.307	2.207	(41)	
(42)		Ebn at Noon	588	733	799	841	857	867	858	852	808	729	598	509	(42)	
(43)		Edh at Noon	62	79	101	116	119	114	112	100	87	71	57	52	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.58	1.26	2.35	3.78	4.80	4.88	4.54	3.73	2.67	1.51	0.73	0.43	(44)	
(45)		RadStd	0.06	0.12	0.22	0.34	0.42	0.49	0.39	0.30	0.15	0.14	0.08	0.04	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page